BookSphere v2.0

Advance Library Management System

Complete Technical Architecture & Features Specification

Prepared for K.R. Mangalam University (KRMU)
Date: 30/5/2026

Ø=ÜŽ Executive Overview

BookSphere v2.0 is an enterprise-grade, high-performance, and secure **Advance Library Management System** engineered for modern educational institutions (pre-configured for K.R. Mangalam University - KRMU).

This document provides an exhaustive, **A-to-Z technical architecture specification and feature inventory**. It serves as an ultimate guide to the system's design, engineering capabilities, security protocols, and operational parameters—entirely free of sensitive credentials.

Ø>Ýí Comprehensive Feature Index (A to Z)

```
mindmap
  root((BookSphere Features))
    Core Operations
      Borrow-Return Lifecycle
      Interactive Wishlists
      Custom Book Reviews
    Role Dashboard
      Owner Panel
      Admin Workspace
      Teacher Access
      Student Portal
    Advanced Modules
      Gemini AI Chatbot
      QR Code Engine
      Peer-to-Peer Exchange
      Reservation Waitlists
      Digital E-Book Suite
    Gamification
      Reading Streaks
      Milestone Badges
      Point Rewards
```

Ø=Üe 1. Role-Adaptive Architecture (RBAC)

BookSphere implements a strict **Role-Based Access Control (RBAC)** model where the user interface dynamically alters its menus, actions, and dashboard stats based on the client's role:

- **Ø=ÜQ Owner (Top Tier):** Holds absolute governance over the entire library system. Can perform all Admin functions plus create/manage Admin accounts, view financial stats, audit logs, and approve/reject administrative registrations.
- **Ø=Þàþ Admin (Staff Tier):** Manages day-to-day operations. Can add/modify/delete books, handle requests, issue/return/renew physical books, view leaderboard ranks, track overdue fines, and manage Student/Teacher accounts.
- **Ø=Üi Ø<ßë Teacher (Privileged User):** Enjoys extended borrow limits and reservation priorities. Can search/reserve books, read eBooks, manage a custom wishlist, submit reviews, and chat with the AI assistant.
- **Ø=Üh Ø<ß“ Student (General User):** Accesses standard library features. Can request books, search catalog, manage personal notifications, track their gamified reading streaks/points, swap books with peers, and download PDF eBooks.

Ø=ÜÚ 2. Core Book Operations

Ø=Ý The Borrow-Return Lifecycle

• **Manual/QR Checkout:** Admin can checkout books manually or via a single scan of the physical book's pre-generated QR code.

• **Request Approval System:** Students can request a book from their dashboard. An Admin reviews the request, clicks "Approve" (which freezes the book copy for 24 hours), and completes checkout when the student collects it.

• **Auto-Expiry Guard:** Approved requests that are not physically picked up within 24 hours automatically expire. The copy is instantly unfrozen and returned to the active inventory, and a notification is sent to the student.

• **Smart Renewals:** Allows students/teachers to extend their return deadline by 7 days, provided the book has no active reservations or waitlist queues.

Ø=ÜÍ Advanced Inventory & QR Code Engine

• **Dynamic Copies Mapping:** Every entry in the catalog maps to distinct physical copies (`BookCopy` schema).

• **Granular QR Codes:** The backend automatically generates clean, base64-encoded DataURI QR codes for each physical copy (e.g., encoded as `BOOKSPHERE:<BookID>:COPY:<CopyNumber>`).

• **Shelf Localization:** Copies are automatically assigned spatial coordinates (e.g., Aisle `A2`, Rack `R4`, Position `P3`) for fast staff physical retrieval.

• **Asset Health Tracking:** Tracks the physical condition of every copy (`new`, `good`, `fair`, `poor`, `damaged`) with a historical log of reported damages and who reported them.

Ø=Þ€ 3. Next-Generation Interaction Modules

Ø=Ü¬ Gemini AI Library Assistant

• Integrated directly inside the front-end chat interface using the high-performance `@google/genai` API.

• **Intelligent Prompting:** Powered by a customized system prompt that loads catalog context, current availability, and active library rules.

• **Capabilities:** Provides natural language recommendations, catalogs search assistance, answers policy questions (e.g., "what is the fine rate?"), and assists with educational research questions.

• **Offline Resilience:** If the API key is not configured, the assistant gracefully falls back to an offline mode, still providing catalog stats and clear directions to contact the Administrator.

Ø=Üe Peer-to-Peer (P2P) Book Exchange

• Allows students and teachers to trade books directly, reducing administrative strain.

• **Workflow:** A user initiates an exchange, selecting a book they currently have and specifying the target recipient. The recipient accepts/rejects the trade. Once accepted, the transaction transfers legally to the new borrower upon Admin validation.

Ø<ßŸþ Active Waitlist / Reservation System

• If a highly popular book has **zero** available copies, users can join a FIFO (First-In, First-Out) waitlist queue.

• **Auto-Notification:** When a copy is checked back in, the system checks the queue, automatically transitions the #1 reserved person's status to `notified`, reserves the copy exclusively for them, and gives them 48 hours to claim it before transferring it to the next person.

Ø<ßÆ Gamified Reader Platform

• Designed to drive higher educational engagement using modern gamification systems:

• **Reading Streaks:** Tracks consecutive borrow dates and rewards active reading behavior.

• **Point System:** Earns points for returning books on-time, writing quality reviews, and maintaining streaks.

• **Badge Milestones:** Earns custom profile badges (e.g., *Bookworm*, *Streak Master*, *Perfect Return*) displayed on the account dashboard.

Ø=Þáþ 9-Layer Modern Security Blueprint

BookSphere's architecture includes a robust, layered security system directly inside `server.js` to safeguard user data and ensure service availability under heavy workloads:

```
[Incoming Request]
        %
        % % %  'v Trust Proxy Config (Clouflare/Nginx IP mapping)
        % % %  'w GZIP Compression (Payload reduction)
        % % %  'x Strict Body Limits (Max 10KB JSON limits)
        % % %  'y CORS Whitelists (Allowed origin lockdowns)
        % % %  'z HTTP Parameter Pollution (HPP parameter guard)
        % % %  '{ Sanitizers (MongoSanitize & XSS sanitization)
        % % %  '| IP Strike Tracker (Blocks scanning bots after 50 errors)
        % % %  '} Timeout Guards (Aborts CPU-hanging queries after 30s)
        % % %  '~ Multi-Tier Rate Limiters (Global / Write / Auth Speeder)
```

1. **'v Trust Proxy Configuration:** Uses `app.set("trust proxy", 1)` to accurately determine client IP addresses behind reverse proxies (like Cloudflare, Nginx, or Heroku), preventing IP spoofing.

2. **'w GZIP Compression:** Compresses response bodies on the fly using `compression()`, saving significant server bandwidth and accelerating page loads for mobile users.

3. **'x Strict Body Size Limits:** Limits incoming JSON payloads to **10 KB** (`express.json({ limit: "10kb" })`) to prevent server buffer overflows or memory-exhaustion attacks.

4. **'y CORS Whitelists:** Enforces a strict, environment-controlled Cross-Origin Resource Sharing (CORS) header. The API rejects requests coming from unauthorized web pages.

5. **'z HTTP Parameter Pollution (HPP):** Employs `hpp()` middleware to protect against parameter pollution attacks (e.g., query strings like `?role=admin&role=student` that can bypass naive input checks).

6. **'{ Database & XSS Sanitizers:**

• `mongoSanitize()`: Automatically strips out dangerous query operators (such as `$gt` or `$ne`) from incoming requests, preventing NoSQL injection attacks.

• `xss-clean`: Sanitizes user inputs to neutralize malicious HTML/JS payloads, blocking Cross-Site Scripting (XSS) in reviews or comments.

7. **'| IP Strike Tracker (Automatic IPS Firewall):** An in-memory monitor that counts client errors (4xx codes, excluding 404s). If an IP generates **50+ errors in a 10-minute window**, it is flagged as a malicious scanner and automatically blocked with a `429 Too Many Requests` code.

8. **'} CPU Request Timeout Guard:** Sets a strict **30-second request timeout limit** on all connections. If a complex DB query hangs or a ReDoS (Regular Expression DoS) attempt occurs, the socket is safely severed to preserve system resources.

9. **'~ Multi-Tier Rate & Speed Limiters:**

• *Global API Limit:* Restricts every IP to a maximum of 300 requests per 15 minutes.

• *Write Mutation Limit:* Restricts state-changing operations (`POST`, `PUT`, `DELETE`) to a maximum of 50 requests per minute.

• *Auth Speed-Limiter:* Dynamically slows down repeated auth requests. If an IP makes more than 5 authentication attempts in 60 seconds, each subsequent request is delayed by an additional **500ms** to render automated brute-force attacks useless.

Ø>Ýê Input Schema & Strict Validation Rules

To prevent data corruption and ensure clean records, BookSphere implements strict schema validation using **Joi** before database entry:

• **Password Complexity Rules:**

• Must be at least **8 characters** long.

• Must contain at least **one uppercase letter**.

• Must contain at least **one lowercase letter**.

• Must contain at least **one number**.

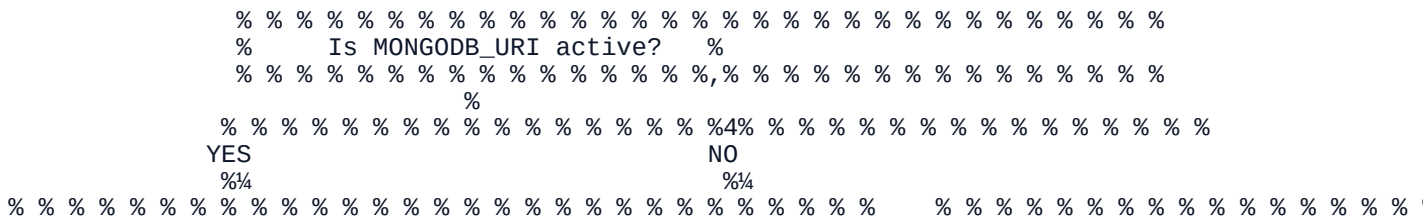
• Must contain at least **one special character** (from `!@#$%^&*`).

• **Domain Registration Restricton:**

• Any registering Student, Teacher, or Admin must use an official college domain email address (ends with `@krmu.edu.in`). *Owner* registrations are exempted from this domain restriction to permit system bootstrap configuration.

Ø=Þâþ Database Strategy & Capacity Specs

Ø=Ý Dual-Engine Strategy



```
%  Connect to MongoDB Atlas  %    %  Start Local Memory Server %
%  (AWS Cloud Hosting Cluster)%   %  (Disk Persistent Fallback)%
% % % % % % % % % % % % % % % % % % % % % % % % % % % % % % %    % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % %
```

1. **MongoDB Atlas (Production Standard):** Connects to a high-availability cloud database hosted on AWS with built-in indexing and continuous optimization.

2. **Persistent WiredTiger Local Fallback:** If the local server lacks internet access or MongoDB Atlas is offline, BookSphere fires up an embedded `mongodb-memory-server` in the background. It is configured to run the advanced **WiredTiger Storage Engine** pointing to `./backend/data/db`.

    • *Note: Unlike standard in-memory engines, this fallback fully preserves your database tables across server restarts and machine reboots!*

Ø=Üæ Storage & Scale Projections

Based on average document sizes (approx. **0.5 KB** per entry), the system can scale as follows:

Database Tier	Allocated Size	Max Document Capacity	Typical Campus Scope
MongoDB Atlas (Free M0)	**512 MB**	**~1,000,000 documents**	Supports **50,000 books**, **10,000 users**, and **500,000 borrow transactions** comfortably.
Local Persistent Disk (SSD)	**50 GB**	**~100,000,000 documents**	Large enough to hold the complete catalog and transaction history of a metropolitan library system.

#ð Automated Workflows & Background Cron Jobs

BookSphere utilizes a background scheduler (`node-cron`) to run automated system updates without manual administrator intervention:

- **#ð Daily Reminder Cron (`0 8 * * *` - Every day at 8:00 AM):**
- Queries all active transactions (`status: "issued"`).
- **2-Day Warning:** If a book is due in exactly 48 hours, it triggers a warning notification and email.
- **Due Today Warning:** If the due date is today, it triggers an urgent return alert.
- **Daily Overdue Escalation:** If the book is overdue, it calculates the dynamic fine (¹5 per day), adds it to the user's account, and fires a daily overdue warning.
- **#ð Approved Request Expiry Cron (`*/10 * * * *` - Every 10 Minutes):**
- Scans the `Request` collection for items approved by an Admin but not claimed within **24 hours**.
- Exhausts the request status to `expired`, **automatically increments the book's available copy count**, and sends a notification informing the user that their hold was released.

Ø<ß¨ Frontend Design Systems & Aesthetics

• **Premium Visuals:** Curated HSL color palette emphasizing deep slate primary tones, smooth turquoise accents, and glassmorphism elements.

• **Modern Typography:** Styled with the geometric Google Font *Inter*, removing generic browser serif styling.

• **Fluid Transitions:** Integrates micro-animations, layout scaling, hover state shifts, and custom animated checkmarks on successful actions.

• **Responsive layouts:** Uses pure Vanilla CSS flexboxes and grid systems to ensure dashboard components scale gracefully from desktop monitors to mobile smartphone screens.

• **Live Charts & Reports:** Uses clean interactive DOM charts on the Admin Dashboard to visualize real-time checkout metrics, category breakdowns, and popular books.

Ø>Ýê Comprehensive Integrity & Testing Framework

BookSphere is supported by a customized local test execution engine ([test-auth-apis.js](file:///c:/Users/ravin/OneDrive/Documents/college/Sem2/Minar%20Project%20-1/Library%20management%20Main/backend/test-auth-apis.js)) checking **62 targeted assertions** to verify system stability:

```
Ø<ß‰ PASS: 62 / 62 Tests Succeeded (100% Success Rate)
```

Ø=Ý, Test Categories Verified:

1. **Auth & OTP Routines:** Validate 6-digit OTP delivery, TTL lifetime expiry, and secure registration checks.
2. **Schema Bounds:** Ensure robust errors are returned for weak passwords, invalid email structures, or unregistered role options.
3. **Role Access Control (RBAC):** Assert that Students or Teachers are strictly blocked (`403 Forbidden`) from viewing logs, approving requests, blocking members, or changing administrative statuses.
4. **Account Management Lifecycle:** Test approving requests, blocking/unblocking accounts, and verify that notifications are correctly written to the collection on state changes.
5. **Rate & Speed Limiter Integrity:** Verify that repetitive fast calls are correctly choked and delayed to protect the server CPU.